

A HIGH INTENSITY SHOCK MYCROCYCLE BOOSTS PERFORMANCE!

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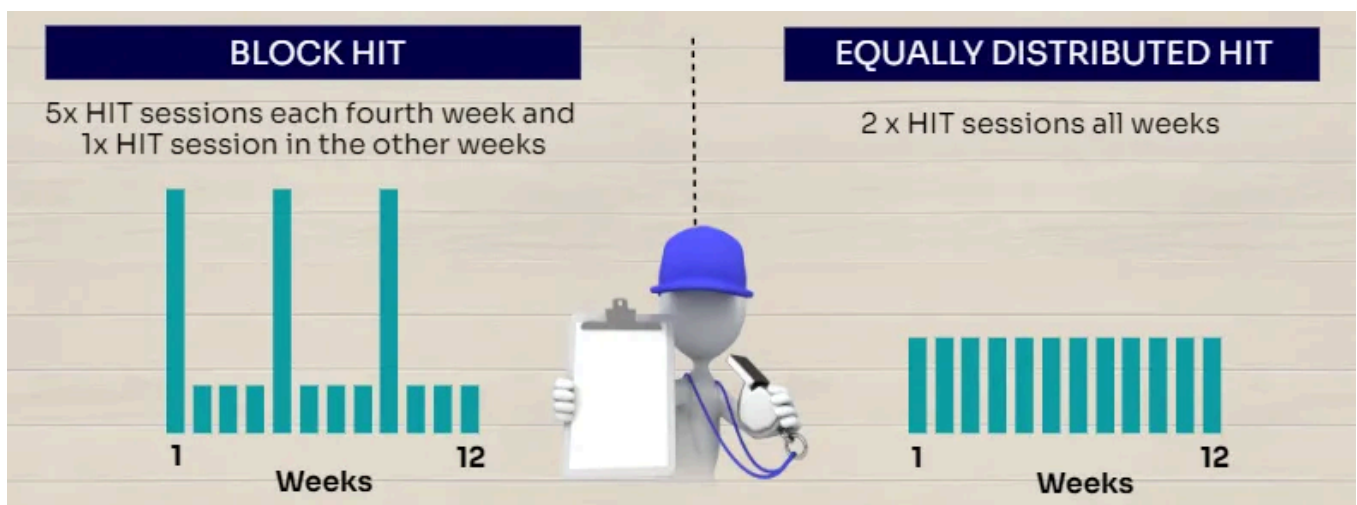
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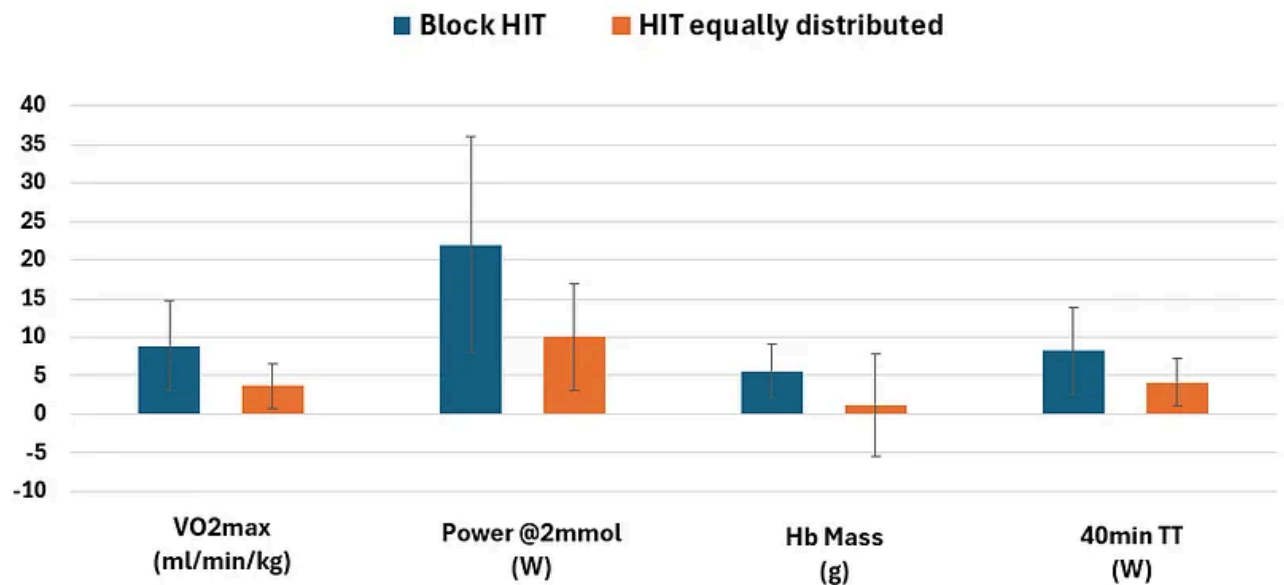
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When considering the annual training plan, linear classic periodization (a progressive increase in volume first and intensity then while keeping or slightly reducing the volume) is the most used strategy by successful endurance cyclists. However, for time-crunched cyclists this solution is not always feasible. In fact, they can usually train only ~1 hour from Monday to Friday, with the possibility to go for long rides only on some weekends. So, they lack time to progressively increase volume and build the linear classic periodization. As a consequence, many usually insert a relevant amount of high intensity training (HIT, intensity above second threshold/FTP/critical power) early in the periodization. In this way, they soon reach a plateau in volume and intensity that is usually kept for several months without progressive overloading the body.

In a [previous KisW research note](#), we saw that another effective solution for (but not limited to) time-crunched cyclists to not maintaining a very similar volume and intensity for several weeks (which could lead to staleness in performance level) is **block HIT periodization**. In the block HIT periodization investigated in the above cited link/study, instead of doing the classic recommended 2/3 times HIT workouts per week for several consecutive weeks (HIT equally distributed), a group of cyclists performed 5 HIT session per week every fourth week, with only one HIT session in the other weeks. The results? They improved more their aerobic parameters and performance compared to the group that performed HIT equally distributed through the weeks.





% changes after 12 weeks of training in the two different groups. Data from Rønnestad et al. (1)

Why HIT Block Periodization is effective? We don't have the definitive answer but it is probably linked to the fact that a more concentrated stimulus on the body can generate a greater adaptive response. However, HIT put a high level of strain on the body, so freshness is crucial to enter a HIT block in order to be capable of absorb the load. So, it is not that easy to perform a HIT block once per month during all the preparation/season. A good solution can be performing it only in the last part of preparation, just before the goal race, during tapering.

Does only one HIT block (HIT crash/shock mycrocycle) just few days before the goal race (during tapering) can optimize performance on race day?

A hot off the press study (2025) published by Solli and Colleagues (Inland Norway University of Applied Sciences, Lillehammer, Norway) on *European Journal of Sport Science* tried to answer this question. (1)

WHAT DID THEY DO?

- 38 well-trained cyclists (34 male and 4 female, age 20, vo2max 70) participated in this study.

- For 12 days, they were divided into two different training groups:
 1. **HIT Shock Mycrocycle (HSM):** performed 5x HIT sessions on the first 6 days (they only rested on the 4th day) and then performed an active recovery period for the next 6 days which consisted in max 1h30min, with just on two days 2x5min@Z3 and 2x1min @Z4/Z5 in a 5/7 zones model

The HIT sessions consisted of what seemed to be the most effective HIT workout to induce aerobic adaptations, the short intermittent intervals (read more [here](#)): 5x 8min45sec alternating 30-s at 118% of 40-min maximal sustainable power output and 15sec at 60% of 40-min PO, with 3min recovery between series.

Compared to the last 2 weeks before entering the study, this group lowered training volume from ~11/13 to ~8 hours per week.

2. **Regular Training (REG):**

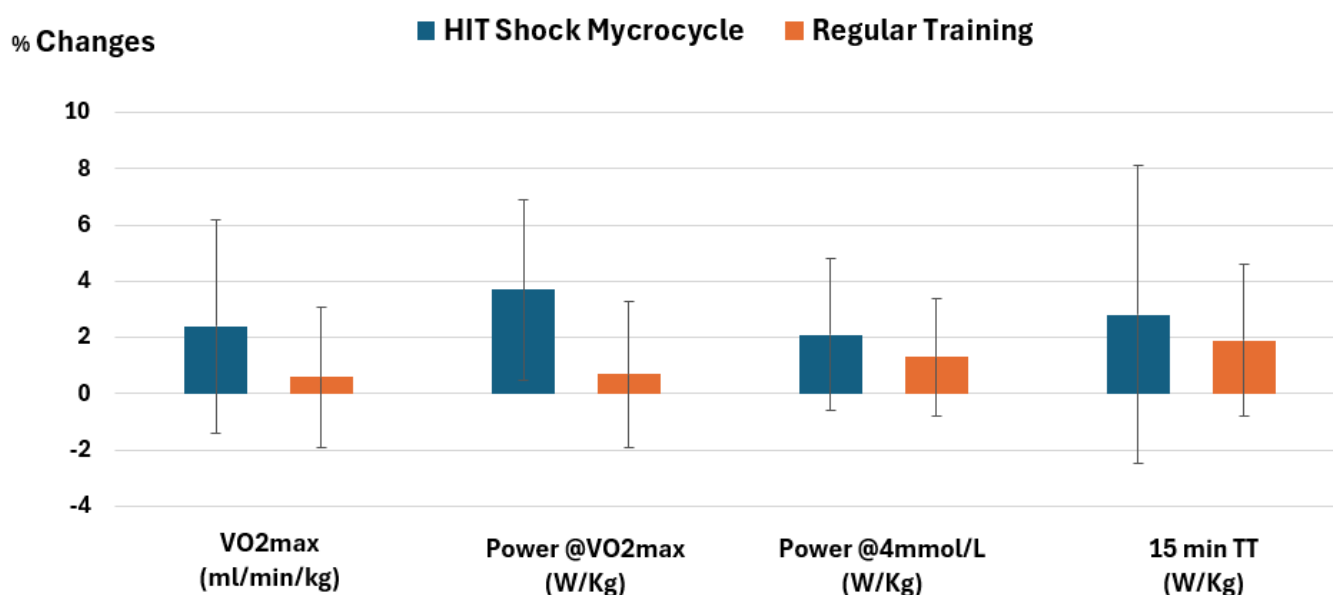
They just continued with their normal training they were doing the last 2 weeks before entering the study, which consisted of performing 2xweek HIT sessions and a training volume of ~11/12 hours per week.

The last two days before testing, both groups perform a similar schedule consisting of a rest day on day -2 and an activation day on day -1.

WHAT DID THEY FIND?

The physiological aerobic parameters and endurance performance increased or tended to increase more in in the Hit Shock Mycrocycle compared to the Regular Training Group:

- **VO2max** (ml/min/kg): $+2.4 \pm 3.8$ vs $+0.6 \pm 2.5$ %
- **Power at VO2max** (W/Kg): $+3.7 \pm 3.2$ vs $+0.7 \pm 2.6$ %
- **Power at 4 mmol/L of Blood Lactate** (W/Kg): $+2.1 \pm 2.7$ vs $+1.3 \pm 2.1$ %
- **Power on a 15min TT** (W/Kg): $+2.8 \pm 5.3$ vs $+1.9 \pm 2.7$ %



% changes after the 12 days of training. Data from Solli et al. (2025)

PRACTICAL TAKEAWAYS:

- A 6 days HIT Shock Mycrocycle followed by 6 days active recovery is effective to improve cycling performance
- This can be used particularly as a strategy on the final taper before the goal race (?).

GABRIELE'S COMMENT

- I think this data can be particularly interesting, especially for time crunched cyclists. They can suggest that, when near to goal race, doing a 5-7 days block of high intensity followed by 5-7 days of just active recovery can be effective to give a final boost to your performance level.
- This strategy can be used also by cyclists with higher training load and not only by time-crunched cyclist. After having already increased first volume and then intensity through the periodization, HIT block periodization could be used at the end of periodization (1 or two blocks), just before the goal race/s to get a final boost. This is very similar to what riders contending grand tours usually do when entering one-week stage races (which represent a huge increase in high intensity volume) after the winter preparatory period.
- The HIT Shock Mycrocycle can be repeated also more than one time during the year, in addition to perform it during the tapering. In this scenario, as HIT training put a high level of strain on the body, before entering a HIT block, it should be given particular attention to make sure to be fresh and rested so that the body can 'absorb' the block.

Why HIT Block Periodization/HIT shock mycrocycle boosts performance?

As stated above, we don't have the definitive answer but it is probably linked to the fact that a more concentrated stimulus on the body can generate a greater adaptive response. This is just a speculation, but if we believe in this principle, we can play around it and create some more alternative training strategies than just what this study used. We will see them in the next posts with some special guests!

Thanks for Reading! Share with your Team!

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1. Rønnestad BR, Ellefsen S, Nygaard H, Zacharoff EE, Vikmoen O, Hansen J, Hallén J. Effects of 12 weeks of block periodization on performance and performance indices in well-trained cyclists. Scand J Med Sci Sports. 2014 Apr;24(2):327-35.
2. Solli GS, Odden I, Sælen V, Hansen J, Mølmen KS, Rønnestad BR. A microcycle of high-intensity short-interval sessions induces improvements in indicators of endurance performance compared to regular training. Eur J Sport Sci. 2025 Jan;25(1):e12223.



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